

13 Method of Manufacture

Agar is obtained by freeze-drying a mucilage derived from *Gelidium amansii* Lamouroux, other species of the same family (Gelidiaceae), or other red algae (Rhodophyta).

14 Safety

Agar is widely used in food applications and has been used in oral and topical pharmaceutical applications. It is generally regarded as relatively nontoxic and nonirritant when used as an excipient.

LD₅₀ (hamster, oral): 6.1 g/kg⁽¹⁰⁾

LD₅₀ (mouse, oral): 16.0 g/kg

LD₅₀ (rabbit, oral): 5.8 g/kg

LD₅₀ (rat, oral): 11.0 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled. When heated to decomposition, agar emits acrid smoke and fumes.

16 Regulatory Status

GRAS listed. Accepted for use as a food additive in Europe. Included in the FDA Inactive Ingredients Database (oral tablets). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral medicines licensed in the UK.

17 Related Substances**18 Comments**

The drug release mechanism of agar spherules of felodipine has been studied and found to follow Higuchi kinetics.⁽¹¹⁾ Agar has also been used to test the bioadhesion potential of various polymers.⁽¹²⁾

The EINECS number for agar is 232-658-1.

19 Specific References

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20 General References**21 Author**

VK Gupta.

22 Date of Revision

10 February 2009.



Albumin

1 Nonproprietary Names

BP: Albumin Solution

PhEur: Human Albumin Solution

USP: Albumin Human

2 Synonyms

Alba; *Albuconn*; *Albuminar*; albumin human solution; albumini humani solutio; *Albumisol*; *Albuspan*; *Albutein*; *Buminate*; human serum albumin; normal human serum albumin; *Octalbin*; *Plasbumin*; plasma albumin; *Pro-Bumin*; *Proserum*; *Zenalb*.

3 Chemical Name and CAS Registry Number

Serum albumin [9048-49-1]

4 Empirical Formula and Molecular Weight

Human serum albumin has a molecular weight of about 66 500 and is a single polypeptide chain consisting of 585 amino acids.

Characteristic features are a single tryptophan residue, a relatively low content of methionine (6 residues), and a large number of cysteine (17) and of charged amino acid residues of aspartic acid (36), glutamic acid (61), lysine (59), and arginine (23).

5 Structural Formula

Primary structure Human albumin is a single polypeptide chain of 585 amino acids and contains seven disulfide bridges.

Secondary structure Human albumin is known to have a secondary structure that is about 55% α -helix. The remaining 45% is believed to be divided among turns, disordered, and β structures.⁽¹⁾

Albumin is the only major plasma protein that does not contain carbohydrate constituents. Assays of crystalline albumin show less than one sugar residue per molecule.

6 Functional Category

Stabilizing agent; therapeutic agent.

7 Applications in Pharmaceutical Formulation or Technology

Albumin is primarily used as an excipient in parenteral pharmaceutical formulations, where it is used as a stabilizing agent for formulations containing proteins and enzymes.⁽²⁾ Albumin has also been used to prepare microspheres and microcapsules for experimental drug-delivery systems.^(3–6)

As a stabilizing agent, albumin has been employed in protein formulations at concentrations as low as 0.003%, although concentrations of 1–5% are commonly used. Albumin has also been used as a cosolvent⁽⁷⁾ for parenteral drugs, as a cryoprotectant during lyophilization,^(8,9) and to prevent adsorption of other proteins to surfaces.

Therapeutically, albumin solutions have been used parenterally for plasma volume replacement and to treat severe acute albumin loss. However, the benefits of using albumin in such applications in critically ill patients has been questioned.⁽¹⁰⁾

8 Description

The USP 32 describes albumin human as a sterile nonpyrogenic preparation of serum albumin obtained from healthy human donors; see Section 13. It is available as a solution containing 4, 5, 20, or 25 g of serum albumin in 100 mL of solution, with not less than 96% of the total protein content as albumin. The solution contains no added antimicrobial preservative but may contain sodium acetyltryptophanate with or without sodium caprylate as a stabilizing agent.

The PhEur 6.0 similarly describes albumin solution as an aqueous solution of protein obtained from human plasma; see Section 13. It is available as a concentrated solution containing 150–250 g/L of total protein or as an isotonic solution containing 35–50 g/L of total protein. Not less than 95% of the total protein content is albumin. A suitable stabilizer against the effects of heat, such as sodium caprylate (sodium octanoate) or *N*-acetyltryptophan or a combination of these two at a suitable concentration, may be added, but no antimicrobial preservative is added.

Aqueous albumin solutions are slightly viscous and range in color from almost colorless to amber depending upon the protein concentration. In the solid state, albumin appears as brownish amorphous lumps, scales, or powder.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for albumin.

Test	PhEur 6.0	USP 32
Identification	+	—
Characters	+	—
Production	+	—
Protein composition	+	—
Molecular size distribution	+	—
Heat stability	—	+
pH (10 g/L solution)	6.7–7.3	+
Potassium	≤ 0.05 mmol/g	—
Sodium	≤ 160 mmol/L	130–160 mEq/L
Heme	+	+
Aluminum	≤ 200 µg/L	—
Sterility	+	+
Hepatitis B surface antigen	—	+
Pyrogens	+	+
Total protein	95–105%	≥ 96%
for 4 g in 100 mL	—	93.75–106.25%
for 5 to 25 g in 100 mL	—	94.0–106.0%
Prekallikrein activator	≤ 35 IU/mL	—

10 Typical Properties

Acidity/alkalinity pH = 6.7–7.3 for a 1% w/v solution, in 0.9% w/v sodium chloride solution, at 20°C.

NIR spectra see Figure 1.

Osmolarity A 4–5% w/v aqueous solution is isoosmotic with serum.

Solubility Freely soluble in dilute salt solutions and water. Aqueous solutions containing 40% w/v albumin can be readily prepared at pH 7.4. The high net charge of the peptide contributes to its solubility in aqueous media. The seven disulfide bridges contribute to its chemical and spatial conformation. At physiological pH, albumin has a net electrostatic charge of about –17. Aqueous albumin solutions are slightly viscous and range in color from almost colorless to amber depending on the protein concentration.

11 Stability and Storage Conditions

Albumin is a protein and is therefore susceptible to chemical degradation and denaturation by exposure to extremes of pH, high salt concentrations, heat, enzymes, organic solvents, and other chemical agents.

Albumin solutions should be protected from light and stored at a temperature of 2–25°C or as indicated on the label.

12 Incompatibilities

See Section 11.

13 Method of Manufacture

Albumin human (USP 32) Albumin human is a sterile nonpyrogenic preparation of serum albumin that is obtained by fractionating material (source blood, plasma, serum, or placenta) from healthy human donors. The source material is tested for the absence of hepatitis B surface antigen. It is made by a process that yields a product safe for intravenous use.

Human albumin solution (PhEur 6.0) Human albumin solution is an aqueous solution of protein obtained from plasma. Separation of the albumin is carried out under controlled conditions so that the final product contains not less than 95% albumin. Human albumin solution is prepared as a concentrated solution containing 150–250 g/L of total protein or as an isotonic solution containing 35–50 g/L of total protein. A suitable stabilizer against the effects of heat such as sodium caprylate (sodium octanoate) or *N*-acetyltryptophan or a combination of these two at a suitable concentration, may be added, but no antimicrobial preservative is added at any stage during preparation. The solution is passed through a bacteria-retentive filter and distributed aseptically into sterile containers,

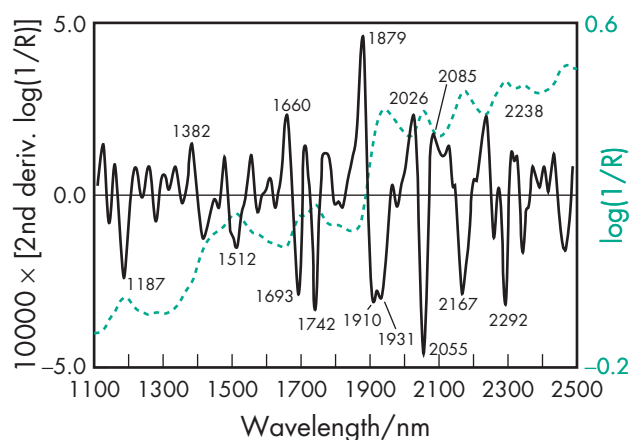


Figure 1: Near-infrared spectrum of albumin measured by reflectance.

which are then closed so as to prevent contamination. The solution in its final container is heated to $60 \pm 1.0^\circ\text{C}$ and maintained at this temperature for not less than 10 hours. The containers are then incubated at $30\text{--}32^\circ\text{C}$ for not less than 14 days or at $20\text{--}25^\circ\text{C}$ for not less than 4 weeks and examined visually for evidence of microbial contamination.

14 Safety

Albumin occurs naturally in the body, comprising about 60% of all the plasma proteins. As an excipient, albumin is used primarily in parenteral formulations and is generally regarded as an essentially nontoxic and nonirritant material. Adverse reactions to albumin infusion rarely occur but include nausea, vomiting, increased salivation, chills, and febrile reactions. Urticaria and skin rash have been reported. Allergic reactions, including anaphylactic shock, can occur. Albumin infusions are contraindicated in patients with severe anemia or cardiac failure. Albumin solutions with aluminum content of less than $200\text{ }\mu\text{g/L}$ should be used in dialysis patients and premature infants.⁽¹¹⁾

LD₅₀ (monkey, IV): $>12.5\text{ g/kg}$ ⁽¹²⁾

LD₅₀ (rat, IV): $>12.5\text{ g/kg}$

15 Handling Precautions

Observe handling precautions appropriate for a biologically derived blood product.

16 Regulatory Acceptance

Included in the FDA Inactive Ingredients Database (oral, tablets, film-coatings; IV injections, IV infusions and subcutaneous injectables). Included in parenteral products licensed in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.

17 Related Substances

Albumins derived from animal sources are also commercially available, e.g. bovine serum albumin.

18 Comments

A 100 mL aqueous solution of albumin containing 25 g of serum albumin is osmotically equivalent to 500 mL of normal human plasma.

The EINECS number for albumin is 310-127-6.

19 Specific References

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20 General References

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21 Author

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22 Date of Revision

12 February 2009.

1 Nonproprietary Names

BP: Ethanol (96%)

JP: Ethanol

PhEur: Ethanol (96 per cent)

USP: Alcohol

2 Synonyms

Ethanolum (96 per centum); ethyl alcohol; ethyl hydroxide; grain alcohol; methyl carbinol.

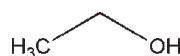
3 Chemical Name and CAS Registry Number

Ethanol [64-17-5]

4 Empirical Formula and Molecular Weight

C₂H₆O 46.07

5 Structural Formula



6 Functional Category

Antimicrobial preservative; disinfectant; skin penetrant; solvent.

7 Applications in Pharmaceutical Formulation or Technology

Ethanol and aqueous ethanol solutions of various concentrations (see Sections 8 and 17) are widely used in pharmaceutical formulations and cosmetics; see Table I. Although ethanol is primarily used as a solvent, it is also employed as a disinfectant, and in solutions as an antimicrobial preservative.^(1,2) Topical ethanol solutions are used in the development of transdermal drug delivery systems as penetration enhancers.⁽³⁻¹⁰⁾ Ethanol has also been used in the development of transdermal preparations as a co-surfactant.⁽¹¹⁻¹³⁾

Table I: Uses of alcohol.

Use	Concentration (% v/v)
Antimicrobial preservative	≥ 10
Disinfectant	60–90
Extracting solvent in galenical manufacture	Up to 85
Solvent in film coating	Variable
Solvent in injectable solutions	Variable
Solvent in oral liquids	Variable
Solvent in topical products	60–90

8 Description

In the BP 2009, the term ‘ethanol’ used without other qualification refers to ethanol containing ≥99.5% v/v of C₂H₆O. The term ‘alcohol’, without other qualification, refers to ethanol 95.1–96.9% v/v. Where other strengths are intended, the term ‘alcohol’ or ‘ethanol’ is used, followed by the statement of the strength.

In the PhEur 6.0, anhydrous ethanol contains not less than 99.5% v/v of C₂H₆O at 20°C. The term ethanol (96%) is used to describe the material containing water and 95.1–96.9% v/v of C₂H₆O at 20°C.

In the USP 32, the term ‘dehydrated alcohol’ refers to ethanol ≥99.5% v/v. The term ‘alcohol’ without other qualification refers to ethanol 94.9–96.0% v/v.

In the JP XV, ethanol (alcohol) contains 95.1–96.9% v/v (by specific gravity) of C₂H₆O at 15°C.

In the *Handbook of Pharmaceutical Excipients*, the term ‘alcohol’ is used for either ethanol 95% v/v or ethanol 96% v/v.

Alcohol is a clear, colorless, mobile, and volatile liquid with a slight, characteristic odor and burning taste.

See also Section 17.

9 Pharmacopeial Specifications

See Table II. See also Sections 17 and 18.

Table II: Pharmacopeial specifications for alcohol.

Test	JP XV	PhEur 6.0	USP 32
Identification	+	+	+
Characters	—	+	—
Specific gravity	0.809–0.816	0.805–0.812	0.812–0.816
Acidity or alkalinity	+	+	+
Clarity and color of solution	+	+	+
Nonvolatile residue	≤2.5 mg	≤25 ppm	≤2.5 mg
Volatile impurities	+	+	+
Absorbance	+	+	+
at 240 nm	≤0.40	≤0.40	≤0.40
at 250–260 nm	≤0.30	≤0.30	≤0.30
at 270–340 nm	≤0.10	≤0.10	≤0.10
Assay	95.1–96.9%	95.1–96.9%	92.3–93.8% by weight 94.9–96.0% by volume

10 Typical Properties

Antimicrobial activity Ethanol is bactericidal in aqueous mixtures at concentrations between 60% and 95% v/v; the optimum concentration is generally considered to be 70% v/v. Antimicrobial activity is enhanced in the presence of edetic acid or edetate salts.⁽¹⁾ Ethanol is inactivated in the presence of nonionic surfactants and is ineffective against bacterial spores.

Boiling point 78.15°C

Flammability Readily flammable, burning with a blue, smokeless flame.

Flash point 14°C (closed cup)

NIR spectra see Figures 1 and 2.

Solubility Miscible with chloroform, ether, glycerin, and water (with rise of temperature and contraction of volume).

Specific gravity 0.8119–0.8139 at 20°C

Note The above typical properties are for alcohol (ethanol 95% or 96% v/v). See Section 17 for typical properties of dehydrated alcohol.

11 Stability and Storage Conditions

Aqueous ethanol solutions may be sterilized by autoclaving or by filtration and should be stored in airtight containers, in a cool place.

12 Incompatibilities

In acidic conditions, ethanol solutions may react vigorously with oxidizing materials. Mixtures with alkali may darken in color owing to a reaction with residual amounts of aldehyde. Organic